

Q.1. Let $A \in M_{n \times n}(\mathbb{C})$ be a rank one matrix. Show that either A is diagonalizable, or A is nilpotent, un both.

Q.2. Give an example of two 4×4 nilpotent matrices which have same minimal polynomial but which are un similar.

Q.3. Let A be a $n \times n$ matrix such that $A^k = 0$ for some positive integer k . Show that $A^n = 0$.

Q.4. Let $T: M_{n \times n}(\mathbb{F}) \rightarrow M_{n \times n}(\mathbb{F})$ be defined by

$B \mapsto AB$
for some fixed $A \in M_{n \times n}(\mathbb{F})$. Is it true that T and A have same eigen values?

Show that the minimal polynomial for T is the minimal polynomial for A .

Q.5. Find primary decomposition of the matrix

$$\begin{bmatrix} 0 & -1 & -1 \\ 1 & 0 & -1 \\ 1 & 1 & 0 \end{bmatrix}$$

over $\mathbb{R}$.

Q.6. Find the projection E which project $\mathbb{R}^2$ onto the space $\text{Span}\{(1, -1)\}$ and $N(E) = \text{Span}\{(1, 2)\}$.

Q.7. If E is a projection on a vector space V , then show that $(I - E)$ is also a projection.

How $N(E)$ and $R(I-E)$ are related.

Q8. If E is a projection on V , show
that $V = R(E) \oplus N(E)$.